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ECA56 –Advanced Wireless Communication Systems / 5G / 6G Communication Systems

Exp: 7 Analog Beamforming Analysis Using MATLAB

Aim :

To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Procedure :

1. Define the system parameters such as carrier frequency, number of antenna elements, wavelength, antenna spacing, and desired beam-steering angle.
2. Calculate the wavelength and antenna spacing using and generally set .
3. Generate the antenna array and calculate beamforming weights using the steering vector for the desired direction.
4. Compute the array factor over a range of angles and convert the magnitude into dB to obtain the beam pattern.
5. Plot and analyze the beam pattern in MATLAB by observing the main beam direction, beamwidth, and side lobes, and verify beam steering by changing the steering angle

Objective – 1 : Analyze Beamforming Gain (dB)

Program :

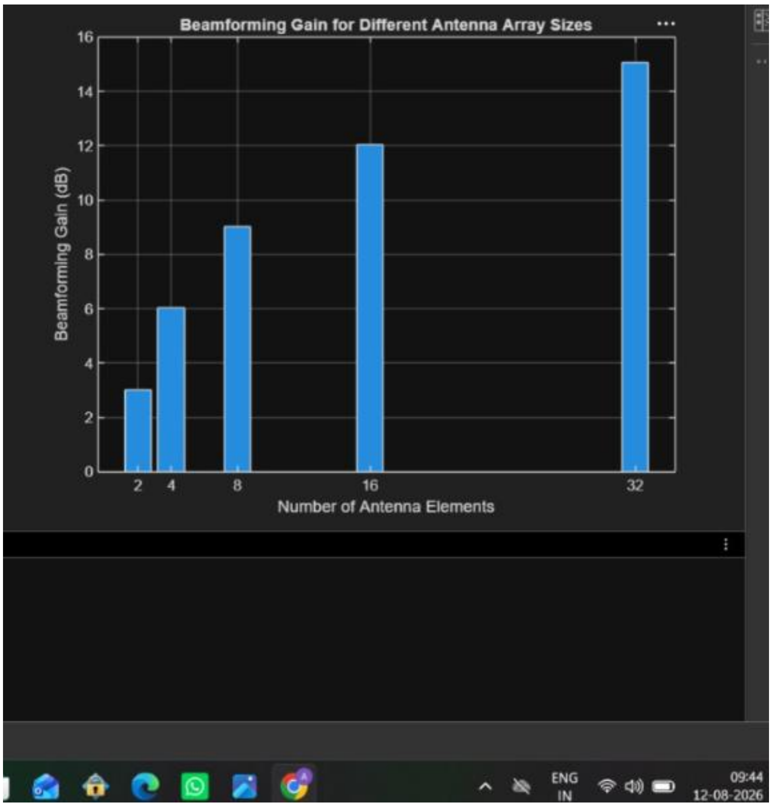
```
clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

gain = 10*log10(antennas); % Beamforming Gain (dB)

figure

bar(antennas, gain)

grid on
```

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

title('Beamforming Gain for Different Antenna Array Sizes')

Output :



Objective – 2 : Compare Beam Width (Degrees)

Program :

clc;

clear;

close all;

antennas = [2 4 8 16 32]; % Number of Antenna Elements

beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)

figure

plot(antennas, beamwidth,'o-','LineWidth',2)

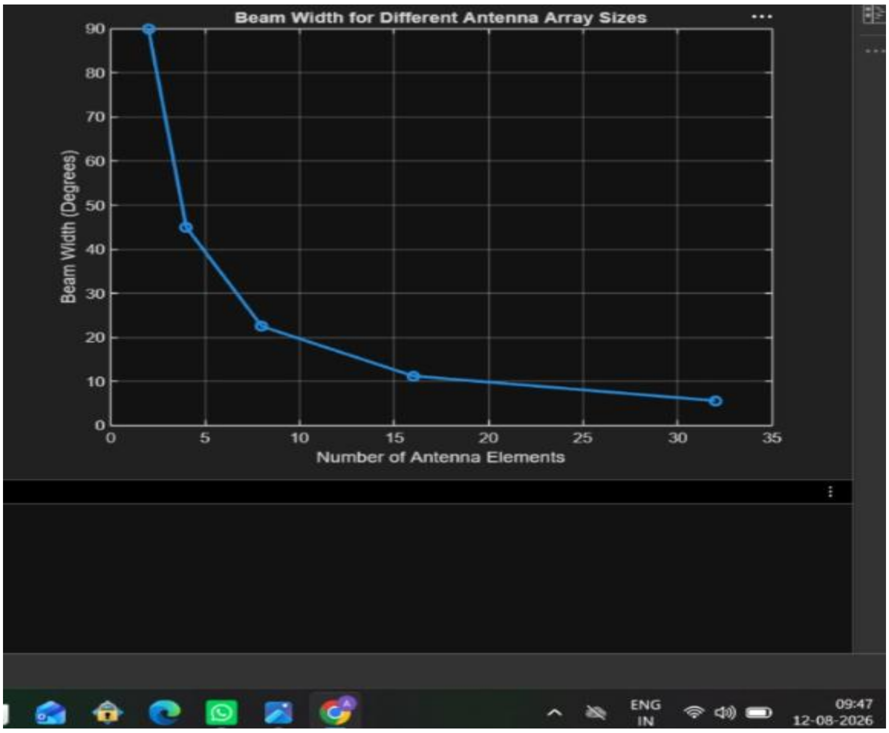
grid on

xlabel('Number of Antenna Elements')

ylabel('Beam Width (Degrees)')

title('Beam Width for Different Antenna Array Sizes')

Output :



Objective – 3 : Analyze Main Lobe Direction (Degrees)

Program :

clc;

clear;

close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)

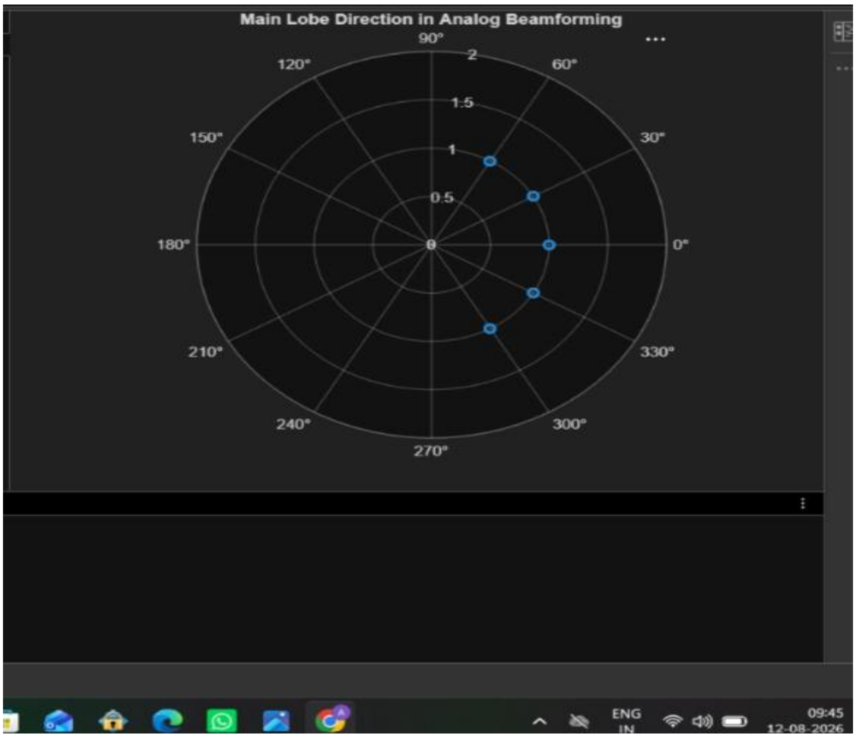
gain = [1 1 1 1 1]; % Constant normalized gain

figure

polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)

title('Main Lobe Direction in Analog Beamforming')

Output :



Result :

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.